

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

SATURDAY, 13 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

Section A (Answer ALL Questions)

- Q1. (a) List **FIVE (5)** inline-level elements. (5 marks)
- (b) Explain the difference between inline-level elements and block-level elements. (4 marks)
- (c) Write the HyperText Markup Language (HTML) to produce the output as shown in **Figure 1**.

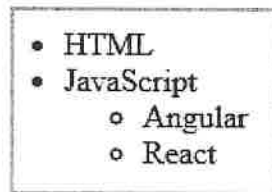


Figure 1: Nested list.

(6 marks)

- (d) The website for “Smart Tuition Center” has the directory structure as shown in **Figure 2**.

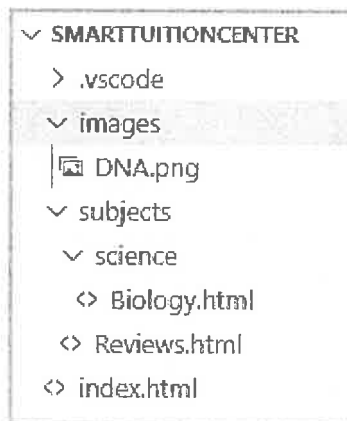


Figure 2: Directory structure.

- (i) Write the HTML markup for “index.html” to create the page as shown in **Figure 3**.

**UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT**

Q1. (d) (Continued)

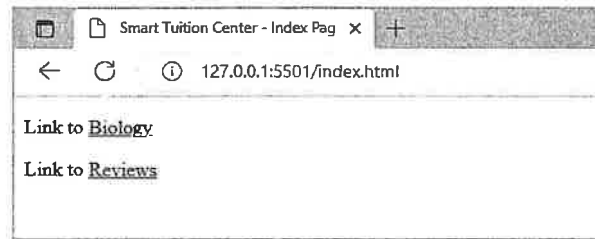


Figure 3: index.html

Note:

- The title bar of the web browser shows “Smart Tuition Center – Index Page”.
- There is a link that links to “Biology.html” page and a link that links to “Review.html” page.

(5 marks)

- (ii) Write the HTML markup for Biology.html page to create the page as shown in **Figure 4**. The web page shows the image “DNA.png”.

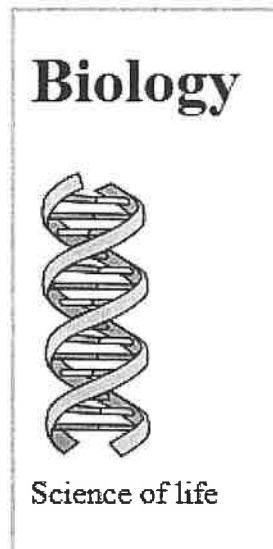


Figure 4: Biology.html

Note:

- Use level 1 heading for “Biology”.

(5 marks)

[Total : 25 marks]

**UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT**

Q2. (a) With the aid of examples, explain the use of **THREE (3)** CSS pseudo-classes. (9 marks)

(b) A web page has the markup as shown in **Figure 5**.

```

26 <body>
27   <h2>Video-sharing Service</h2>
28   <section id="main">
29     <h2>
30       Microsoft Stream
31     </h2>
32     <article>
33       <header>
34         <h2 class="green">Stream (on SharePoint), the new version of Microsoft Stream</h2>
35       </header>
36       <p>
37         Empowers you to record, upload, discover, share, and manage video just as you would any other
38         file.
39       </p>
40     </article>
41     <article>
42       <header>
43         <h2>Vimeo - Video Experience Platform</h2>
44       </header>
45       <p>
46         Vimeo, Inc. is an American video hosting, sharing, and services platform provider headquartered in New
47         York City.
48       </p>
49     </article>
50   </section>
51 </body>

```

Figure 5: Applying embedded CSS

Write the embedded CSS to create the following styles:

- (i) Everything in the second level headings will be italicized. For example, the sentences such as “*Microsoft Stream*”, “*Stream (on SharePoint), the new version of Microsoft Stream*” and “*Vimeo - Video Experience Platform*” will appear in italics in the web browser. (3 marks)
 - (ii) The sentence “Microsoft Stream” in line 30 will appear in red when it is displayed in the web browser. (3 marks)
 - (iii) The sentence “Stream (on SharePoint), the new version of Microsoft Stream” in line 34 will appear in green when it is displayed in the web browser. (3 marks)
 - (iv) The first article will have yellow background color when it is displayed in the web browser. (3 marks)
- (c) (i) What is the advantage of using external styles sheet? (2 marks)

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

Q2. (c) (Continued)

- (ii) Write the HTML markup to link to the external style sheet called “styles.css”. (2 marks)

[Total : 25 marks]

- Q3. (a) Nancy is assigned to develop a webpage to compute the total price of an item. The web page allows the user to input the unit price and the quantity of the item. When the “Calculate” button is clicked, it will display the total price to the user. The sample output is as shown in **Figure 6**. Write the JavaScript code to handle the click event of the “Calculate” button.

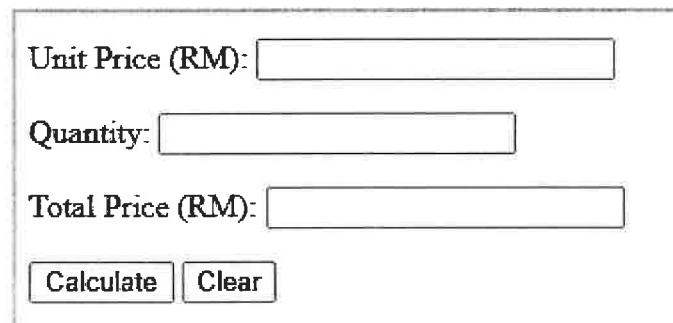


Figure 6: Sample output.

(10 marks)

- (b) List **TWO (2)** ways to find HTML elements. Give **ONE (1)** example for each of them. (6 marks)
- (c) A web page has a text box for user to enter a list of fruits as shown in **Figure 7**. Write the JavaScript code that will obtain the list of fruits from the text box and display each of the fruits in an unordered list in descending order when the “Submit” button is clicked. For example, if the user enters “Banana, Orange, Pineapple, Apple” into the text box, then the output will be as shown in **Figure 8** when the “Submit” button is clicked.

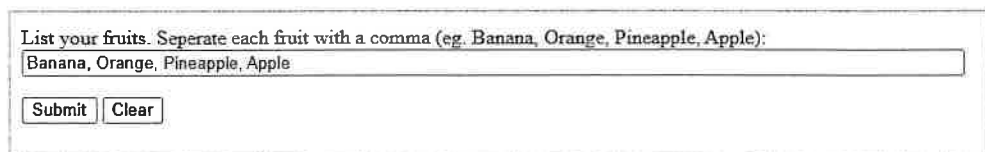
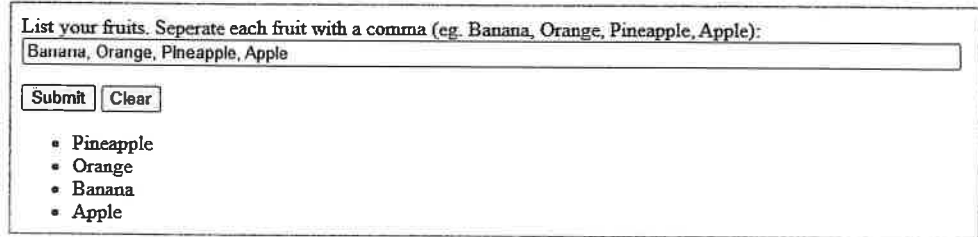


Figure 7: Web page that allows user to enter a list of fruits

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

Q3. (c) (Continued)



The screenshot shows a web form with a text input field containing the text "Banana, Orange, Pineapple, Apple". Above the input field is a label: "List your fruits. Seperate each fruit with a comma (eg. Banana, Orange, Pineapple, Apple):". Below the input field are two buttons: "Submit" and "Clear". Below the buttons is a bulleted list of fruits: "Pineapple", "Orange", "Banana", and "Apple". The fruits are listed in descending order of their position in the input string.

List your fruits. Seperate each fruit with a comma (eg. Banana, Orange, Pineapple, Apple):

Banana, Orange, Pineapple, Apple

Submit Clear

- Pineapple
- Orange
- Banana
- Apple

Figure 8: Fruits are displayed in descending order.

(9 marks)
[Total : 25 marks]

**UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT**

Section B (Answer only ONE (1) question)

Q4. (a) Compare the **THREE (3)** mechanism for storing data on users' computer using JavaScript. (9 marks)

(b) Write the HTML markup and JavaScript code to create the web page as shown in **Figure 9**. When the "Save" button is clicked, the background color entered into the text box will be saved into local storage and the background color of the web page will change to the background color entered into the text box. However, if the browser does not support web storage, then an alert message with the message "Sorry, your browser does not support Web Storage..." will be shown and the background color of the web page will remain.



Figure 9: Background colour preference.

(11 marks)

(c) With the aid of markup, explain how caption can be incorporated into a video. There will be two captions, "English" and "Chinese" captions. The "Chinese" caption will be the default. (5 marks)

[Total : 25 marks]

Q5. (a) (i) A famous blogger wants to keep track on the number of times her blog has been visited by the users. The blog page will display the number of times that the page has been visited as shown in **Figure 10**. Propose a programming technique to achieve the above criteria. Support your proposal with the HTML markup and JavaScript code.



Figure 10: Blog page

(12 marks)

(ii) Give **ONE (1)** drawback of the proposed technique in Q5(a)(i)? (5 marks)

**UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT**

Q5. (Continued)

- (b) A web developer is presented with the markup as shown in **Figure 11**. Write the JavaScript to display the value of the “fruitsId” in an alert message box when a fruit is clicked. The sample output when “Orange” is clicked is as shown in **Figure 12**.

```

1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="utf-8">
6      <title>Q5 (b)</title>
7      <script src="js/jquery-3.6.3.min.js"></script>
8  </head>
9
10 <body>
11     <div>
12         <p data-fruitsid="1">Apple</p>
13         <p data-fruitsid="2">Orange</p>
14         <p data-fruitsid="3">Banana</p>
15     </div>
16 </body>
17 </html>

```

Figure 11: q5b.html

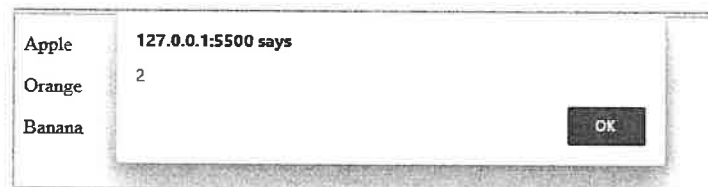


Figure 12: Sample output when "Orange" is clicked.

(4 marks)

- (c) List and describe the **TWO (2)** jQuery methods that can be used to send and receive data asynchronously from a web server.

(4 marks)

[Total : 25 marks]